

UME414: Solids and Structures

L	T	P	Cr
3	1	2*	4.0

* Lab on the alternate weeks

Course Objective: This subject aims to develop an understanding of the stresses and strains that develop in solid materials when they are subjected to different types of loading and to develop an understanding of the conditions at failure of such materials. Further, this subject aims at to introduce the fundamental concepts of structural mechanics.

Syllabus

Axial Stress and Strain: Concept of stress, strain, elasticity and plasticity; one-dimensional stress-strain relationships; Young's modulus of elasticity, shear modulus and Poisson's ratio; two-dimensional elasticity; isotropic and homogeneous materials; ductile and brittle materials; statically determinate and indeterminate problems, compound and composite bars; thermal stresses. Torsion of shafts; buckling of struts, concept of factor of safety.

Review of Shear Force and Bending Moment Diagrams

Bending & Shear Stresses in beams: Derivation of flexural formula for straight beams, concept of second moment of area, bending stress calculation for beams of simple and built up sections, Flitched beams. Shear stress formula for beams, shear stress distribution in beams

Transformation of Stress and Strain: Transformation equations for plane stress and plane strain, Mohr's stress circle, relation between elastic constants, strain measurements, strain rosettes.

Deformations: Governing differential equation for deflection of straight beams having constant flexural rigidity, Double Integration, Macaulay's and Moment Area and Conjugate Beams for determining slopes and deflection.

Laboratory Work

1. Calculation of tensile strength using UTM
2. To determine the Hardness of various materials
3. Experimental verification of Theory of bending (Calculation of bending stress and deflections at various points in the beam theoretically and verifying the same experimentally) and indirect evaluation of the modulus of elasticity
4. Torsion: Study the behavior of circular shafts under torsion and analysis of failure and indirect evaluation of the modulus of rigidity.
5. Estimation of the impact strength using impact testing.

Experimental project assignment: Students in groups of 4/5 will do projects related to the experiments conducted in the laboratory.

Course Learning Objectives (CLO)

The students will be able to:

1. evaluate axial stresses and strains in various determinate and indeterminate structural systems
2. analyse the circular shafts subjected to torsion
3. to study the effect of combined stresses and strains
4. determine deformations and deflections in various kinds of beams and trusses

Text Books

1. Hibbeler, R.C., Mechanics of Materials, Pearson, 11th impression (2018).
2. Popov, E.P. and Balan, T.A., Engineering Mechanics of Solids, Prentice Hall of India (2012).
3. Rattan, S.S., Strength of Materials, McGraw Hill Education (2017).

Reference Books

1. Shames, I. H. and Pitarresi, J. M., Solid Mechanics, Prentice Hall of India (1996).
2. Crandall, S.H., Dahl, N.C. and Lardner, T.J., An Introduction to Mechanics of Solids, McGraw Hill International, Tokyo (1969).

Evaluation Scheme

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	25
2.	EST	35
3.	Sessionals (May include Assignments/Projects/Tutorials/Quiz/Lab evaluations)	40